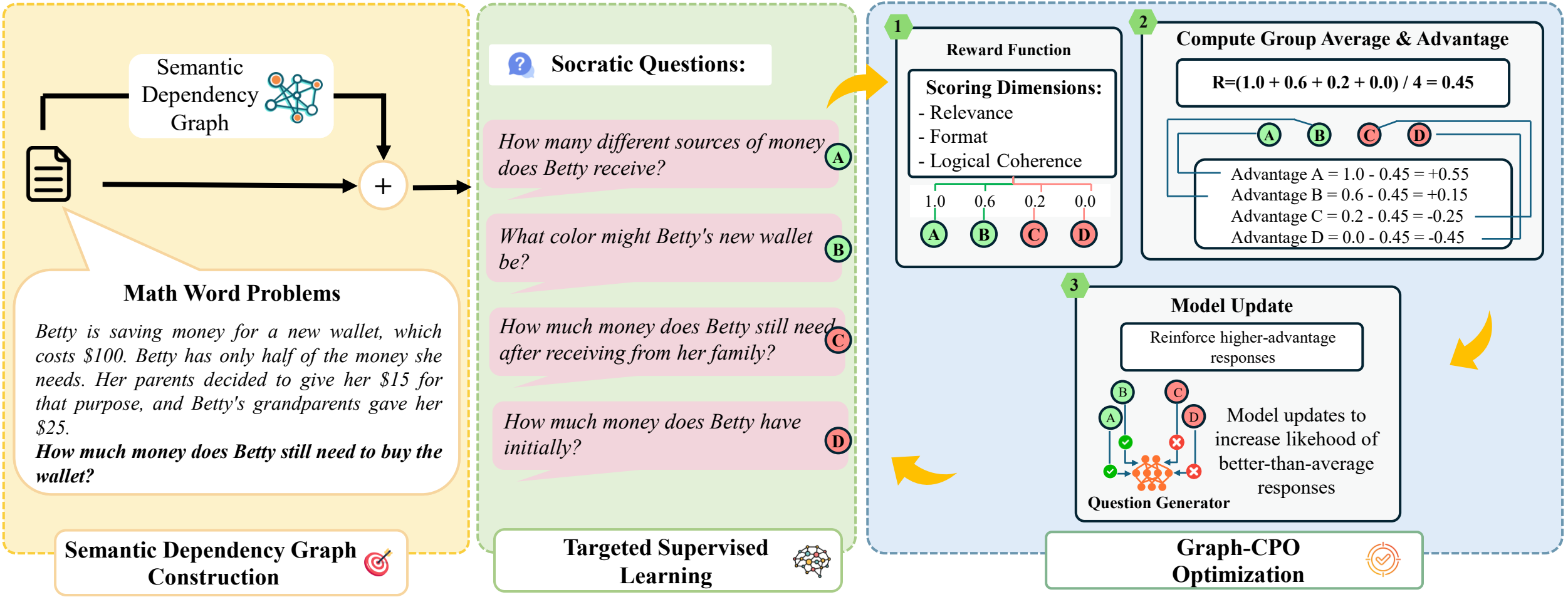

Semantic Dependency Graph
Math Word Problems
Betty is saving money for a new wallet, which costs $100. Betty has only half of the money she needs. Her parents decided to give her $15 for that purpose, and Betty's grandparents gave her $25.
How much money does Betty still need to buy the wallet?
Socratic Questions:
How many different sources of money does Betty receive?
A
What color might Betty's new wallet be?
B
How much money does Betty still need after receiving from her family?
C
How much money does Betty have initially?
D
1
Reward Function
Scoring Dimensions:
- Relevance
- Format
- Logical Coherence
1.0
0.6
0.2
0.0
A
B
C
D
2
Compute Group Average & Advantage
R=(1.0 + 0.6 + 0.2 + 0.0) / 4 = 0.45
A
B
C
D
Advantage A = 1.0 - 0.45 = +0.55
Advantage B = 0.6 - 0.45 = +0.15
Advantage C = 0.2 - 0.45 = -0.25
Advantage D = 0.0 - 0.45 = -0.45
3
Model Update
Reinforce higher-advantage responses
Model updates to increase likehood of better-than-average responses
Question Generator
Semantic Dependency Graph Construction
Targeted Supervised Learning
Graph-CPO Optimization